



Effects of Chinese herbal residues of *Huang Huo oral liquid* coupled with cow dung on the growth, reproduction and gene expression of earthworms

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ABSTRACT

This study aims to solve the problem of environmental pollution caused by dung and herbal residues left after the extraction of the effective components of Chinese herbal medicine. Herbal residues of *Huang Huo oral liquid*, *Astragalus membranaceus* and *Epimedium* coupled with cow dung were used in breeding earthworms. Earthworm growth and reproduction indices were recorded and transcriptome analysis of differentially expressed genes was carried out between earthworms in control group and herbal residues groups. The results showed that herbal residues promoted the growth and reproduction performance of earthworms. The differentially expressed genes in earthworms after the addition of herbal residues of *Huang Huo oral liquid* were associated with glucose fat and protein metabolic pathways, AMPK signaling pathway and oxytocin signaling pathway. In group D (control group), the up-regulated, differentially expressed genes inhibit glycogen, fatty acid and protein synthesis processes as well as cell growth processes. In group A, among the mostly down-regulated differentially expressed genes, the function of genes with transcript numbers Cluster-2829.0 and Cluster-30676.1 relates to the oxytocin signaling pathway in earthworms. The analysis showed that the gene expression of energy metabolism-related signaling pathway and oxytocin signaling pathway in earthworms was changed and the growth and reproduction function of earthworms was enhanced after the addition of 30% herbal residues of *Huang Huo oral liquid*.

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1. Introduction

With the rapid development of animal husbandry, large amount of dung produced in the process of animal farming has become a major issue that restricts animal husbandry development and pollutes environment (Pergola et al., 2018). The European Union proposed as early as in 2012 that, in order to cope with farming-related environmental and climate stresses and resource depletion, the ways of producing, processing, storing, recycling and disposing such biological resources as dung should be effectively changed (European Commission, 2012). Cattle rearing industry is an essential part of animal husbandry, and the accumulation of cow dung leads to emissions of a large amount of greenhouse gas and odorous gas, which has a huge impact on the air (Moller et al., 2014; Santos et al., 2016). In addition, cow dung contains a large amount of nitrogen, phosphorus and potassium. Although these substances can be used as soil improvers to promote plant growth (Nielsen and Hofmann, 2021), the leaching of cow dung during the accumulation process will lead to soil,

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soil nutrients and water pollution. In addition, people's lives and health are threatened by a certain amount of pathogenic microorganisms, drug residues, heavy metals and other contaminants in cow dung (Sun et al., 2016). Improper disposal of these wastes will also have a negative impact on the environment and climate (Nsiah-Gyambibi et al., 2021).

Chinese herbal medicine is an important component in the medical and health care industry (Zhou et al., 2016). In recent years, people's increasing awareness of the dangers of such synthetic drugs as antibiotics has become the reason why the demand for herbal products and their raw materials has multiplied (Charuau et al., 2019; Su et al., 2020; Li et al., 2022). *Huang Huo oral liquid* is the active ingredients prepared from *Astragalus membranaceus* and *Epimedium*, which have the functions of improving physical strength and enhancing body immunity. It is mainly used in clinical practice to improve the immunity of chickens, to improve the antibody level to vaccination and to prolong the action time of antibodies in the body of birds when used in synergy with vaccines. According to statistics, China produces at least 30 million tons of herbal residues every year (Jin et al., 2020). However, there are still 30~50% of medicinal active ingredients remaining in the residues after the extraction (Li et al., 2017). The random stacking of herbal residues not only pollutes the environment, but also causes a great waste of nutrients and medicinal active ingredients.

Vermicomposting technology emerged as the "yard vermicomposting method" in the United States in the early 1980s. Up to now, vermicomposting is often used to dispose of biological waste (Kausar and Khwairakpam, 2022). Vermicomposting, as a biotechnology, can convert organic wastes into nutrient-rich humus-like substances, reducing or eliminating the pollution of organic wastes to the environment (Nunes et al., 2016; Bhat et al., 2018; Cai et al., 2022). Earthworms alter the composition of waste by reducing organic waste toxicity, organic carbon content, C/N ratio, and ultimately increase macro and micro nutrient levels in the final product (Ahmed and Deka, 2022). Vermicomposting is an excellent technique for processing organic waste.

Vermicomposting has great potential for the treatment of cow dung and herbal residues. It not only makes the environment better protected, but also maximizes the use of resources. In this study, vermicomposting was conducted using dung and herbal residues to explore: (i) Whether vermicomposting solve the problem of environmental pollution caused by dung and herbal residues, (ii) the mechanism of herbal residues affecting the growth and reproduction of earthworms, (iii) the genes and metabolic pathways of herbal residues to promote the growth and reproduction of earthworms using genomics technology.

2. Materials and methods

2.1. Earthworms

Earthworms (*Eisenia fetida*) used in this experiment were purchased from an earthworm breeding base in Shijiazhuang city, China. Mature earthworms with worm rings and weight between 200–400 mg were selected for the experiment. These selected earthworms were acclimatized in the laboratory for one week.

2.2. Reagent preparation

Chloroform, isopropanol, and anhydrous ethanol used in the experiments were purchased from a commercial supplier (Beijing Chemical Factory, China). Liquid nitrogen was obtained from Hengyuan Acetylene Gas Plant, Baoding, China. TRIzol was bought from Thermo Fisher scientific (Shanghai, China). DEPC water was supplied by Sigma-Aldrich (Shanghai, China). Herbal residues of *Huang Huo oral liquid* were provided by Jinkun Pharmaceutical Co., Ltd, Xinji, China. The soil moisture meter is purchased on jd.com shopping platform, product model: 52286585929(Wuhan, China). Oven purchased from YIHENG17 company (Shanghai, China), product model: DHG-9140A. Autoclave purchased from SHENAN company (Shanghai, China), product model: LDZM-40. Centrifuge purchased from Bioridge company (Shanghai, China), product model: TD4.

2.3. Experimental set up

The herbal residues were mixed with fermented cow dung in the ratio of 3:7 (herbal residue is 900 g and the cow dung is 2,100 g) as the substrate for earthworms (Singh and Suthar, 2012). The test was performed in 4 groups, namely A, B, C and D, each in 3 replicates. Groups A was the herbal residues of *Huang Huo oral liquid* group (a mixture residues of *Astragalus membranaceus* 450 g and *Epimedium* 450 g), Group B was the herbal residues of *Astragalus membranaceus* group (residues of *Astragalus membranaceus* 900 g). Group C was the herbal residues of *Epimedium* group (residues of *Epimedium* 900 g). Group D was the blank control group. The substrate in each replicate group weighted 3000 g and was put into a polyethylene container. The substrate was laid flat in the polyethylene container to form an earthworm bedding of 50 cm in length, 32 cm in width and 5 cm in thickness. After the earthworm bedding was fully prepared and hydrated with distilled water, 10 worms were placed on the earthworm bed and their conditions were observed. When the earthworms did not escape and were in good condition, 200 g of earthworms of the same age were placed in each polyethylene container of groups A, B, C and D. At the same time, cow dung mixed with herbal residues in the ratio of 7:3 was laid on the surface, and then sprayed with distilled water. In group D, only cow dung was added. The treated polyethylene containers were placed outdoors for 32 days for the earthworms to grow and reproduce and distilled water was sprayed onto them every two days to maintain the moisture of the breeding beds at 60%–70% (use soil moisture meter). Sampling was conducted every 8 days and 20 earthworms were collected from each replicate group.

2.4. Growth and reproduction index determination

On days 0, 8, 16, 24 and 32, 20 earthworms were randomly taken from each replicate group, weighed and the average daily gain of earthworms was calculated. Then the earthworms were placed at 4 °C for 5–10 min to achieve the effect of anesthesia. All anesthetized earthworms were kept in the same posture to facilitate the measurement of their body length.

The number of earthworm cocoons after the 32-day test were counted and the cocoons were picked out to hatch in the same environment. The hatching time and hatching rate of earthworm cocoons were recorded.

2.5. RNA extraction

On Day 32, nine earthworms in groups A and D were taken out, washed with distilled water, and then placed in a petri dish covered with filter paper. The filter paper was moistened with distilled water for 48 h without eating or drinking. The earthworms were then anesthetized with distilled water at 4 °C. Liquid nitrogen was quickly added after the earthworms were placed on a foam box, and the earthworms were taken 30 s later for anatomy on ice. Then the head of the earthworm containing various organs was excised from the posterior part of the ring band and the abdomen of the earthworm was dissected longitudinally. The earthworm with the intestinal contents removed was rinsed with anhydrous ethanol to prevent contamination, and the obtained tissues were stored at –80 °C.

The taken earthworm tissues were placed in a mortar for fully grinding in 1mL TRIZOL, and then the homogenate was transferred to a 2mL enzyme-free centrifuge tube and centrifuged at 12,000 rpm at 4 °C for 10 min. The supernatant was transferred to a 1.5mL centrifuge tube and vigorously shaken for 15 s. 200 µL chloroform was added, evenly mixing was conducted upside down and then the tube was left at room temperature for 5 min. It was centrifuged at 12,000 rpm at 4 °C for 15 min. The supernatant was transferred into a 1.5mL tube, an equal volume of isopropanol pre-cooled at 4 °C was added. Gently mixing was done and then the tube was left at room temperature for 10 min. It was centrifuged at 12,000 rpm at 4 °C for 15 min, the upper layer of liquid was discarded, 1mL of RNase-free 80% ethanol was added to wash the precipitate. And then it was centrifuged at 10,000 rpm at 4 °C for 5 min. The supernatant was discarded, the centrifuge tube was placed upside down on the toilet paper, ventilation and drying were done for 5–10 min, and then 20 µL of DEPC water was added to wait for RNA to be completely precipitated into it. After the precise integrity detection of the 2 µL of extracted RNA using Agilent 2,100 bioanalyzer, cDNA library was established.

2.6. Statistics analysis

The weight and length of earthworms was performed using one-way analysis of variance (ANOVA) with SPSS statistical software (Version19.0), $p < 0.05$ was significantly different, and $p < 0.01$ was extremely different.

All cDNA samples were sequenced on the high-throughput sequencing platform (Illumina), and the measured image data were transformed by CASAVA base identification into sequence data (reads), which are called Raw Reads in Fastq format and mainly contain the sequence information of sequenced fragments and their corresponding sequencing quality information. Then the quality test and filter were conducted to Raw Reads to obtain the final data used for analysis (Clean Reads). The quality of the sequencing data was reflected by checking the distribution of sequencing error rates at individual base positions, i.e. Raw Read quality test, where the sequencing error rate is represented by Error and the sequencing base quality by Qphred, with the relationship $Qphred = -10 \log_{10} e$. Then Raw Reads, which contain a small amount of reads with sequencing adapters or of low sequencing quality, were filtered, and reads with adapters were removed. Reads with N (N means base information cannot be determined) were removed. Then reads of low quality (reads with Qphred ≤ 20 bases accounting for more than 50% of the entire read length) were removed to finally obtain Clean Reads.

For transcriptome sequencing without reference genes, Clean Reads were spliced to obtain reference sequences for subsequent analysis. Trinity (Haas et al., 2013) was used in the process. After removing the redundancy of transcript sequences obtained by Trinity splicing using Corest hierarchical clustering, the longest Cluster sequence (Unigene) was obtained, and then functional annotation was performed through comparison with five databases, namely Nr, Nt, GO, KOG/COG and KEGG.

The transcripts obtained after Trinity splicing were used as reference sequences (Ref), and the Clean Reads of each sample were mapped towards the Ref, filtering out the reads with alignment quality values lower than 10, reads failing to pass pairwise alignment, and reads that were aligned to multiple regions of the genome, respectively. RSEM software was used for alignment (Li and Dewey, 2011; Gamini et al., 2021), and the bowtie2 parameter used in RSEM was mismatch0 (the default parameter of bowtie2). RSEM (Li and Dewey, 2011) counted the bowtie alignment results for subsequent gene expression level analysis. The number of reads aligned to each gene from each sample was called readcount. The readcount was converted into FPKM value, and the intra- and inter-group correlation coefficients were calculated based on the FPKM values of all genes in the samples using Pearson correlation coefficients. The closer to 1 the correlation coefficient is, the higher the similarity of expression patterns between samples is. We required the square (R^2) of Pearson correlation coefficient to be at least greater than 0.8.

The readcount obtained from the gene expression level analysis was used as the data for differential gene screening. For samples with biological replicates, the DESeq (Nagy et al., 2021; Simon and Wolfgang, 2010) method was used, with the

Table 1

Weight and length change of earthworms during outdoor vermicomposting with the herbal residues.

Group	Day 0		Day 8		Day 16		Day 24		Day 32	
	Weight (mg)	Length (cm)	Weight (mg)	Length (cm)	Weight (mg)	Length (cm)	Weight (mg)	Length (cm)	Weight (mg)	Length (cm)
A	240.33 ±9.4	5.17 ±0.03	438.17 ±15.56 ^{Aa}	7.09 ±0.28 ^{Aa}	460.5 ±6.08 ^{Aa}	8.50 ±0.23 ^{Aa}	526.33 ±12.60 ^{Aa}	8.00 ±0.1 ^{Aa}	389.67 ±1.74 ^{Aa}	7.13 ±0.09 ^{Aa}
B	273 ±2.52	4.97 ±0.09	329.50 ±26.94 ^{Bb}	6.40 ±0.24 ^{Aba}	359.83 ±10.06 ^{Cc}	6.97 ±0.07 ^{Bb}	496.83 ±11.02 ^{Aa}	7.47 ±0.09 ^{Aab}	361.33 ±3.03 ^{Bb}	6.77 ±0.03 ^{Ab}
C	254.5 ±12.11	5.13 ±0.03	425.16 ±11.35 ^{Aa}	6.72 ±0.24 ^{Aba}	414.17 ±2.62 ^{Bb}	6.79 ±0.53 ^{Bb}	418.17 ±6.74 ^{Bb}	7.17 ±0.35 ^{Ab}	343.50 ±0.87 ^{Cc}	6.15 ±0.03 ^{Bc}
D	255.67 ±14.58	4.97 ±0.13	384.33 ±3.38 ^{ABa}	6.11 ±0.12 ^{Ab}	413.83 ±1.64 ^{Bb}	6.77 ±0.13 ^{Bb}	410.33 ±13.71 ^{Bb}	6.97 ±0.24 ^{Ab}	336.33 ±0.17 ^{Cd}	5.90 ±0.17 ^{Bc}

Notes: In the same column, different capital letters on the shoulder label mean extremely significant difference ($p < 0.01$), different lowercase on shoulder label showed significant difference ($p < 0.05$), and shoulder label with the same letter indicate no significant difference ($p > 0.05$). The Table 2 same as Table 1.

probability of hypothesis testing (p -value) calculated through the negative binomial distribution statistical model. Finally Benjamini–Hochberg (Haynes, 2013; Ghosh, 2019) procedure was performed to obtain the FDR value (False discovery rate, padj is its common form and stands for FDR), while the log2 Fold Change value was calculated using the readcount average, with the absolute value of log2 Fold Change value greater than or equal to 1 and padj less than 0.05 as the differential gene screening criteria. Finally, the differential genes in each group were counted, and the number of differential genes unique and common among the groups were analyzed, and the differential genes finally screened were subjected to GO and KEGG functional enrichment analysis to obtain the functional annotation of the differential genes.

3. Results and discussion

3.1. Effect of herbal residues on the growth and reproduction of earthworms

As shown in Table 1, on the 8th day, earthworms in group B had the lowest average weight, significantly different from all other groups ($P < 0.05$) and extremely different from groups A and C ($P < 0.01$). The average body length of earthworms in group A was significantly longer than that in group D ($P < 0.05$), indicating growth-promoting effect of the herbal residues of the mixture of *Astragalus membranaceus* and *Epimedium*. On the 16th day, the average body weights of earthworms in groups A and B were significantly ($P < 0.01$) greater than those in groups C and D. The average weight of earthworms in group A was the greatest while the average weight of earthworms in group B was the lowest. Earthworms in Group A had the longest body length, significantly different from all other groups ($P < 0.01$), suggesting a growth-promoting effect of the mixed herbal residues. On the 24th day, significant differences were observed in average weights of earthworms between the four groups ($P < 0.01$) with the order from groups A, B, C to D (the highest to the lowest). The body length of earthworms in group A was significantly higher compared with groups C and D ($P < 0.01$). On the 32nd day, the average weight and body length of earthworms were respectively in groups A, B, C, and D from highest to lowest, and the average weight of each group differed significantly ($P < 0.05$). The difference between groups A and B and the other two groups was extremely significant ($P < 0.01$). The average length of earthworms in groups A and B was significantly different from the other two groups ($P < 0.05$). The results showed that the herbal residues of *Huang Huo oral liquid* promoted the growth of earthworms in weight and length during vermicomposting. The residue of *Astragalus membranaceus* promoted the weight gain of earthworms in the later stage of vermicomposting, but not the length growth of earthworms. The residue of *Epimedium* had no significant effect on the growth of earthworms.

As shown in Table 2, after 32 days of vermicomposting with 30% herbal residues added in cow dung, the number of earthworm cocoons was listed from high to low in groups C, A, D and B, and the difference between the groups was highly significant ($P < 0.01$). The earthworm cocoons in group A took longer time to hatch than those in all other groups and the difference was highly significant ($P < 0.01$). There was no significant difference in hatching time of earthworm cocoons between groups C, D and B ($P > 0.05$). The hatching rate of earthworm cocoons in group A was significantly lower than all other groups ($P < 0.01$). The experiment showed that both herbal residues of *Huang Huo oral liquid* and of *Epimedium* could improve the cocoon production of earthworms but had no significant effect on the hatching time and hatching efficiency of earthworm cocoons. 30% residues of *Astragalus membranaceus* had an inhibitory effect on the reproduction of earthworms.

3.2. Effects of herbal residues on gene expression in earthworms

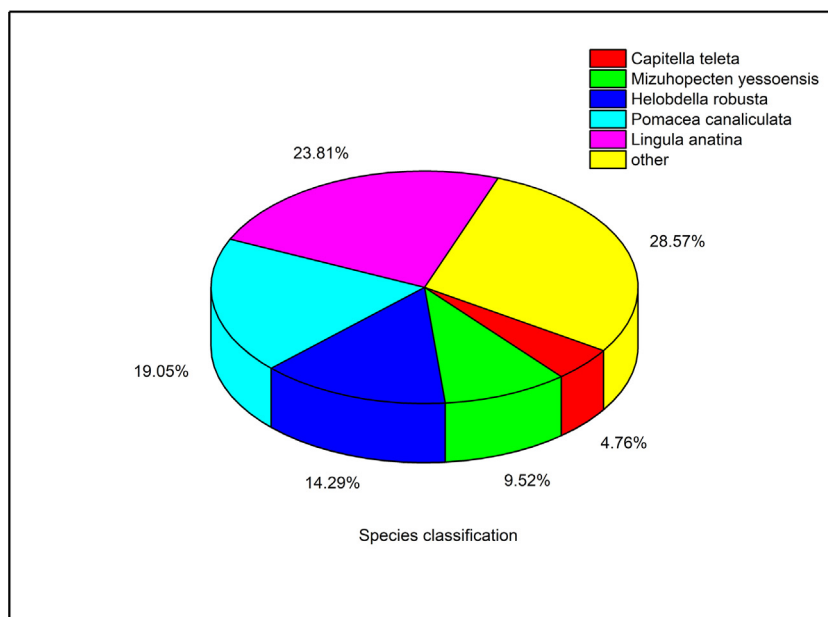
3.2.1. Splicing of transcripts and functional annotation

After the genes obtained after all cDNAs were de-redundantly screened were annotated (details of cDNA screening are placed in supplementary material), as shown in Fig. 1, an alignment annotation between the 43,755 Unigenes obtained

Table 2

Reproduction of earthworms during outdoor vermicomposting with the herbal residues.

Group	Herbal residues	Proportion	Number of earthworm cocoons	Hatching time/d	Hatching rate
A	Huang Huo Oral Liquid	3:7	88 $\pm$ 1.73 ^{Bb}	17.33 $\pm$ 0.33 ^{Bb}	80.00 $\pm$ 2.89 ^{Aa}
B	Astragalus membranaceus	3:7	25.67 $\pm$ 0.88 ^{Dd}	20.33 $\pm$ 0.33 ^{Aa}	66.67 $\pm$ 1.67 ^{Bb}
C	Epimedium	3:7	117 $\pm$ 3.46 ^{Aa}	17 $\pm$ 0.00 ^{Bb}	83.33 $\pm$ 1.67 ^{Aa}
D	–	3:7	58 $\pm$ 3.06 ^{Cc}	17 $\pm$ 0.00 ^{Bb}	81.67 $\pm$ 1.67 ^{Aa}

**Fig. 1.** Similarity of gene sequence between *Eisenia foetida* and related species.

and the Nr database shows the similarity between the gene sequence 3 of *Eisenia foetida* and the gene sequence of closely related species. 16.5% of their gene sequences were similar to *Capitella teleta* among the phylum Annelida, and 15.6% their gene sequences were similar to *Helobdella robusta* of the phylum Annelida. In addition, the proportions of gene sequences similar to *Lingula anatina*, *Mizuhopecten yessoensis* and *Pomacea canaliculata* were 6.9%, 3.0% and 2.7% respectively, and the above animals belong to the phylum Brachiopoda and the phylum Mollusca separately. Earthworms are oligochaetes of the phylum Annelida and have a good gene sequence match with other annelida, indicating that the transcripts are well spliced and there is some match with the gene sequences of the phylum Brachiopoda and the phylum Mollusca.

As shown in Fig. 2, 12,485 among 43,755 Unigenes were successfully aligned with the GO database. Gene encoded products were classified according to the biological processes involved, the molecular functions they have and the cellular component, with 41 groups of functional annotations obtained.

The biological processes that gene encoded products mainly get involved in are intracellular formation and metabolism, chiefly cellular macromolecule metabolism, nitrogen compound metabolism, organocyclic compound metabolism, organonitrogen compound metabolism, heterocyclic metabolism, cellular aromatic compound metabolism, cellular biosynthesis and primary metabolic regulation. The molecular functions possessed are mainly binding and catalysis reactivity, chiefly nucleic acid binding, cation binding, anion binding, transferase activity, ion transmembrane transporter protein activity, etc. Gene encoded products mainly function in cells and protein-containing complexes, especially organelles, membrane-bound organelles, non-membrane organelles, organelle membranes, and membrane protein complexes.

Unigene was aligned with different Ortholog clusters composed of homologous genes from different species in the KOG database, and the results are shown in Fig. 3. Of the 5,746 Unigenes successfully annotated, 14.42% were associated with post-translational modifications, protein conversion and molecular chaperone functions, 11.85% with predicted involvement in general functions, and 11.81% with signal transduction mechanism.

As shown in Fig. 4, after KOG annotation of Unigene genes, the KEGG metabolic pathways were classified according to the metabolic pathways of their products and compounds in cells and the functions of these gene products. The metabolic pathways are classified into five types: a (Cellular Processes), b (Environmental Information Processing), c (Genetic Information Processing), d (Metabolism) and e (Organismal Systems). Among the 6,906 genes successfully annotated,

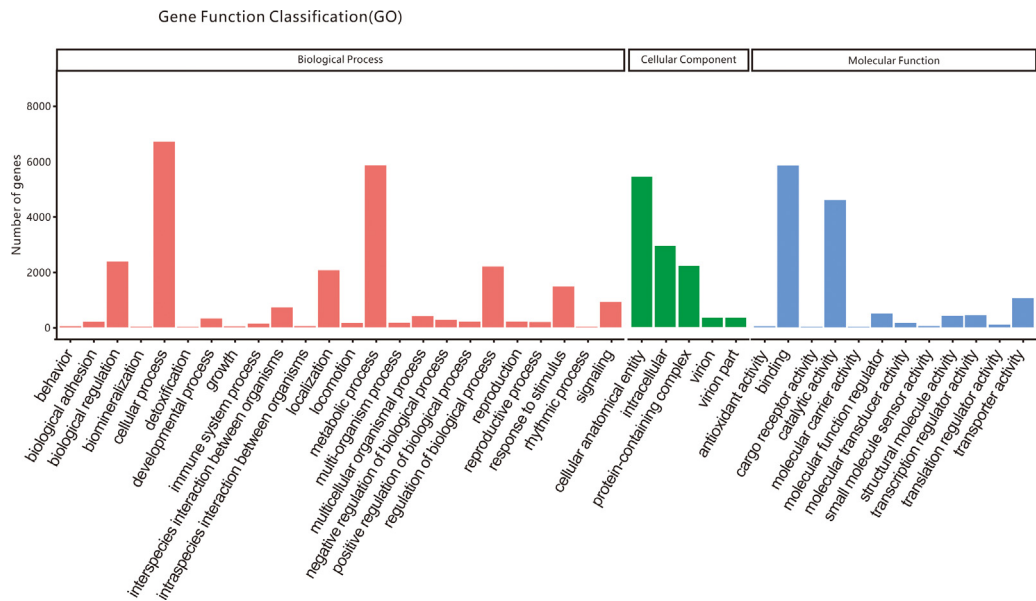


Fig. 2. GO classification of Eisenia foetida Unigene function.

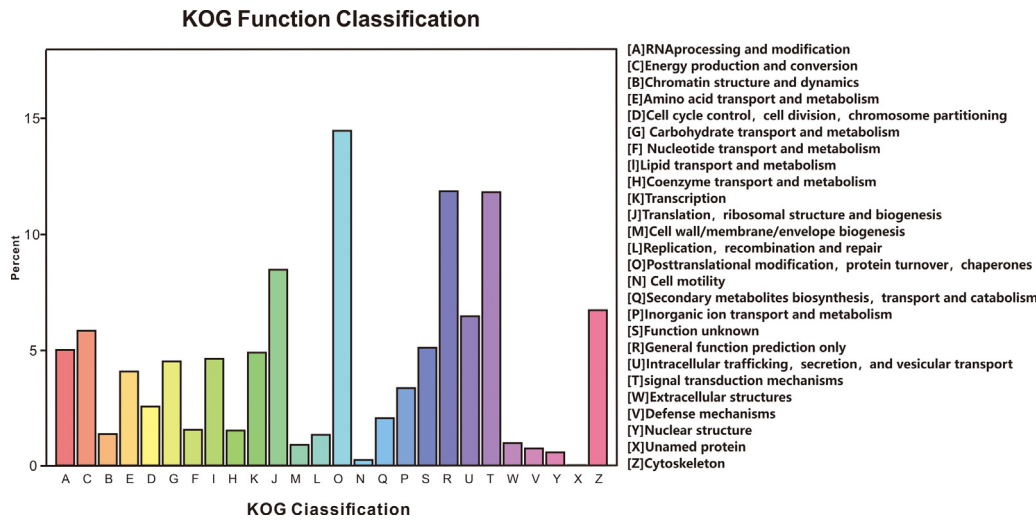


Fig. 3. KOG function annotation of Eisenia foetida Unigene.

735 genes (10.64% of the total) were involved in transport and catabolism in cellular processes, mainly the cell focal adhesion pathway, endocytosis, lysosomal action, tight junction action and actin cytoskeleton regulation; 1,148 genes were involved in signal transduction in environmental information processing, accounting for 16.62% of the total, and the main participating signal transduction pathways were PI3K-Akt signaling pathway, MAPK signaling pathway, Rap1 signaling pathway, cAMP signaling pathway, cGMP-PKG signaling pathway, etc; 537 genes were involved in folding, sorting and degradation of genetic information processing, accounting for 7.77% of the total, mainly including endoplasmic reticulum protein processing, ribosomes, spliceosome and RNA transport, ubiquitin-mediated protein degradation and other processes; 449 genes were involved in the metabolic process of carbohydrate metabolism, accounting for 6.50% of the total, mainly including carbon dioxide metabolism, purine metabolism, oxidative phosphorylation and amino acid biosynthesis processes; 628 genes were involved in the endocrine system of the organic system, accounting for 9.09% of the total, mainly including oxytocin signaling pathway, insulin signaling pathway, adrenaline signaling pathway, estrogen signaling pathway and thyroid hormone signaling pathway. Among the Unigene genes, genes involved in the organic system of the body was in the largest number, totally 2808 genes.

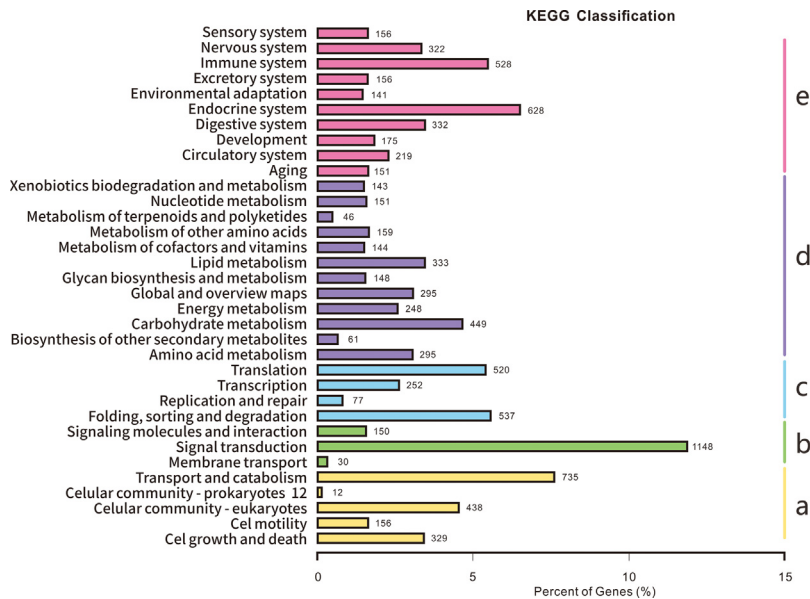


Fig. 4. Functional KEGG classification of Eisenia foetida Unigene.

Table 3
Alignment of Clean Reads with reference genes.

Sample name	Total reads	Total mapped
D1	44060580	36004710(81.72%)
D2	44520184	33753796(75.82%)
D3	41835570	32330796(77.28%)
A1	39344010	30547284(77.64%)
A2	39357274	29681140(75.41%)
A3	39869448	28916672(72.53%)

3.2.2. Gene expression differences and functional analysis

The results are detailed in Table 3. The Clean Reads of each sample were mapped to the reference sequence (Ref) of the spliced transcript, and the alignment rate of the number of reads of the sample to the gene was 72.53–81.72%, which was 60% higher than the alignment rate and satisfied the requirements of the subsequent analysis.

According to the screening criteria that the absolute value of log2 Fold Change is greater than or equal to 1 and *p*-value less than 0.05, the readcount aligned to genes in group D and group A was counted, and a total of 2,326 genes were significantly differentially expressed in group D compared to group A. Among them, 1,108 genes were significantly up-regulated differentially expressed and 1,218 genes significantly down-regulated differentially expressed.

As shown in Table 4 and supplementary material, the genes significantly differentially expressed in group A compared with group D were among the 20 pathways with the highest enrichment in KEGG function, nine of which were directly related to the three major metabolic pathways, including the pentose phosphate metabolic pathway, glycolysis/gluconeogenesis, protein digestion and absorption metabolic pathway, adipocytokine signaling pathway and phosphatidylinositol signaling pathway. Genes in the AMPK signaling pathway, a key pathway associated with biological energy metabolism, were also significantly enriched. As shown in Table 6, genes with elevated expression in the AMPK signaling pathway in group D eventually inhibited glycogen synthesis, fatty acid synthesis, glucagon gene expression, cell growth, protein synthesis, glycolysis and gluconeogenesis.

The results are detailed in supplementary material. The top 10 genes that were significantly regulated differentially expressed in the group A compared with the group D were selected according to log2 Fold Change absolute values in descending order. In group D, the function of the genes with transcript numbers Cluster-2829.0 and Cluster-30676.1 was related to the oxytocin signaling pathway in earthworms. As show in Table 5 Of the 10 genes associated with the oxytocin signaling pathway, six genes were up-regulated in expression in addition to the two genes mentioned above. In addition to regulating oxytocin expression, the down-regulated genes were also associated with protein synthesis and sugar uptake in the body.

Table 4
Genes related to AMPK signaling pathway.

gene_id	log2FoldChange	pval	Expression	Name of the gene to which it belongs	Numbering of notes	Function
Cluster-18315.0	22.164	0.000004998	Up	PIK3CA	K00922	Inhibited cell growth and protein synthesis
Cluster-26610.0	7.8802	0.020692	Up	PRKAG	K07200	Inhibited glycogen synthesis
Cluster-28233.1	8.2716	0.013521	Up	ACACA	K11262	Repressed fatty acid synthesis
Cluster-26055.0	7.1039	0.045378	Up	pckA	K01596	Inhibited expression of glucogenic genes
Cluster-20068.0	22.121	0.000004998	Up	GYS	K00693	Inhibited glycogen synthesis
Cluster-21793.0	8.2192	0.015236	Up	PPP2R2	K04354	PP2A
Cluster-26916.2208	9.5508	0.00075803	Up	pfkA	K00850	Inhibited expression of glucogenic genes

Table 5
Differential genes related to oxytocin signaling pathway.

gene_id	log2FoldChange	Expression	Name of the gene to which it belongs	Numbering of notes	Function
Cluster-6593.0	-7.4215	Down	CALM	K02183	Regulation of OXT signaling
Cluster-30676.1	-10.288	Down	PRKCA	K02677	Protein synthesis
Cluster-2829.0	-10.143	Down	EEF2	K03234	Protein synthesis
Cluster-2159.1	-7.19	Down	EEF2	K03234	Protein synthesis
Cluster-4625.0	-7.4742	Down	PKA	K04345	Inhibition proliferation
Cluster-1308.0	-7.3572	Down	EEF2	K03234	Protein synthesis

3.3. Effect of herbal residues on the growth and reproduction of earthworms

The results of vermicomposting showed that herbal residues of *Huang Huo oral liquid* could promote the growth and reproduction of earthworms, the residues of *Huang Huo oral liquid* is the residue after extracting *Huang Huo oral liquid* from *Astragalus membranaceus* and *Epimedium*. The reason why *Huang Huo oral liquid* promotes the growth of earthworms may be that the components of *Astragalus* play a role. Studies have shown that *Astragalus membranaceus* can increase the growth performance of broilers and piglets, and improve the laying performance of laying hens (Wang et al., 2010, 2020). There may be a large amount of polysaccharide components and active factors in *Astragalus membranaceus* that can promote the growth of animal bodies (Alagawany et al., 2022), and it is these polysaccharide components and active factors that increase the energy intake of earthworms, resulting in the increase of earthworms' weight. During the composting, herbal components and nutrients affect the growth of earthworms (Petersen et al., 2007). In addition, the addition of herbal residues will enrich the cellulose content of the material, improve the palatability of the material, change the number and type of bacteria in the material, and ultimately promote the growth of earthworms, a conclusion with Biruntha et al. (2020) the same conclusion. However, in the late stage of vermicomposting, the body weight of earthworms in each group showed a downward trend, especially in the group of herbal residues of *Astragalus membranaceus*, the body weight of earthworms decreased significantly. This may be due to herbal residues lacking of nutrients for earthworm growth in the later period.

Herbal residues of *Huang Huo oral liquid* group and *Epimedium* group both improved the reproductive performance of earthworms, which was mainly due to the effect of *Epimedium*. *Epimedium* can regulate the level of hormones in the body, inhibit the apoptosis of spermatogenic cells, and improve reproductive performance (Li et al., 2015). This is also the reason why the *Epimedium* group has the best reproductive indicators. According to the test results, the herbal residues had no significant effect on the hatching rate and hatching time of earthworm cocoons. The hatching rate and hatching time of worm cocoons still need further study. In addition, the adequacy of oxygen and pH during vermicomposting can affect the growth of earthworms. It is worth noting that the changes in the weight and length of earthworms are not caused by one factor, and further research is needed.

3.4. Effect of herbal residues on gene expression in earthworms

Gene annotation was conducted after de-redundancy of transcripts obtained from earthworm splicing. It was found through gene annotation that the main signaling pathways in the organic system of earthworms are oxytocin signaling

pathway, insulin signaling pathway, adrenaline signaling pathway, estrogen signaling pathway and thyroid hormone signaling pathway. Oumi et al. (1994) early research found that the body of earthworm contains a kind of neuropeptide that is highly homologous to oxytocin and Oumi named it earthworm oxytocin. Subsequently, the chemical structure of earthworm oxytocin was further revealed in Acher (1993) study, further confirming its presence and function. In Ricketts et al. (2004), it was further demonstrated that oxytocin in earthworms can affect earthworm reproduction and cocoon production as well as the activity of earthworms' crop and gizzard, leading to changes in the digestive function of the earthworm.

The gene expression produced by the earthworms in the herbal residues of *Huang huo oral liquid* group was significantly different from that in the control group. The differentially expressed genes were concentrated in three major metabolic processes. Glucose metabolism, lipid metabolism and protein metabolism are the most basic metabolic processes in living organisms and also necessary metabolic pathways to provide living organisms with energy and maintain their normal physiological functions. When the animal body glucose metabolism, lipid metabolism, protein metabolism processes are enhanced, it can increase the muscle content in the animal body, improve the growth performance of animals and accelerate the synthesis of ATP to provide more energy for the body. In the test, differential expression of genes related to the three major metabolisms occurred, indicating that the enhanced metabolism and the increased energy intake of the earthworm after the addition of the herbal residues of *Huang Huo oral liquid* led to differential growth of earthworms. AMPK signaling pathway is an important pathway for energy metabolism in the body. The up-regulated genes in the herbal residues group were enriched in the AMPK signaling pathway, and the up-regulated genes inhibited glycogen synthesis, fatty acid synthesis, glucose-raising gene expression, cell growth and protein synthesis, glycolysis and gluconeogenesis, indicating that the addition of herbal residues altered the energy metabolic process. In addition, compared to group D, the up-regulated differential expression genes in group A were enriched in oxytocin signaling pathway-related pathways, indicating that the high expression of oxytocin-related genes in earthworms under the addition of herbal residues of *Huang Huo oral liquid* resulted in different expression of oxytocin in earthworms, thus improving their reproductive capacity. The oxytocin signaling pathway can also have an effect on the AMPK signaling pathway, which ultimately affects the energy metabolic process of the body. This suggests that earthworm oxytocin signaling pathway is altered with the addition of herbal residues of *Huang Huo oral liquid*, which affects the expression of earthworm oxytocin and ultimately improves the growth and reproduction of earthworms.

Also, the experiment has some interesting findings. Earthworms have high genetic similarity with *Capitella teleta* and leeches among annelids and certain genetic similarity with some mollusks, which is consistent with the results of previous studies (Simakov et al., 2013; Shah et al., 2000). Studies also pointed out that mollusks and annelids share the same cell division pattern during early development (Henry et al., 2007). This suggests that they are related in species evolution and that there is some functional similarity of genes. There are abundant genetic studies on *Capitella teleta* and leeches, which can be used as a reference for earthworm genetic studies.

In addition to being widely used in human society, Chinese herbal medicine will gradually penetrate into animal husbandry in the future. It is inevitable for us to consider how to deal with the herbal residues. This work is to explore the effect of vermicompost in herbal residues, and to provide valuable experience for future related staff.

4. Conclusion

Herbal residues of *Huanghuo Oral Liquid* changed the gene expression of energy metabolism-related signaling pathways and oxytocin signaling pathways in earthworms, and promoted the growth and reproduction of earthworms, and this mechanism is the differentially expressed genes in earthworms which were associated with glucose fatty acid and protein metabolic pathways, AMPK signaling pathway and oxytocin signaling pathway.

CRedit authorship contribution statement

Zhanfei Ma: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Yang Li:** Conceptualization, Methodology, Software, Writing – review & editing. **Xiaoyong Ma:** Methodology, Investigation, Visualization. **Yabo Wang:** Writing – review & editing. **Jungang Kang:** Writing – review & editing, Resources. **Guojun Jiang:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Research data for this article

The data that support the findings of this study may be available upon reasonable request to the corresponding author.

Appendix A. Supplementary data

Supplementary data related to this article can be found in e-version of this paper online.

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.eti.2022.102893>.

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